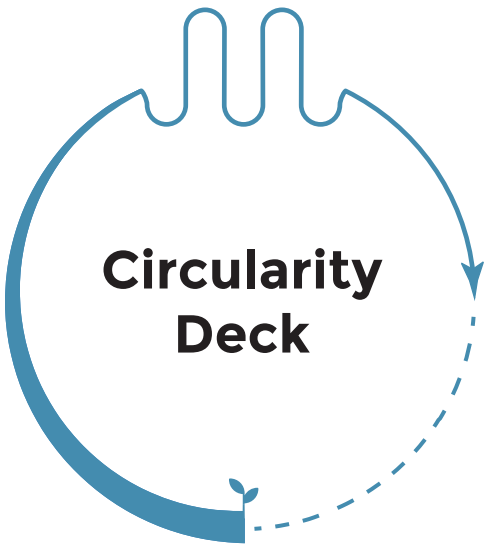




www.circularstrategies.org

TERMS OF USE

The Circularity Deck has been developed by Jan Konietzko during his PhD Research at TU Delft, supervised by Nancy Bocken and Erik Jan Hultink.

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You can order printed versions of the Circularity Deck. For that, please reach out to Jan Konietzko (for example on LinkedIn).

QUICK INTRO (I)

The Circularity Deck is a collection of circular economy principles, organized according to five key strategies that you can apply to the material and energy flows of your business:

- 1) Narrow (use less material and energy during design, production, delivery and use)
- 2) Slow (use products, components and material longer)
- 3) Close (Use wasted products, components and materials again)
- 4) Regenerate (Use non-toxic and biodegradable materials and renewable energy)
- 5) Inform (Use information technology to narrow, slow, close and regenerate material and energy flows)



QUICK INTRO (II)

Each principle may require a product, business model or ecosystem perspective. This is indicated on each card. For example, a *product principle* for 'narrowing' is 'design with low-impact inputs'.

To use the cards, you need a print-out of the circular economy framework. Then go through the cards and map them around the framework. Use post-its to capture: is this being done already? If yes, by whom? How? If not, can this be done? How? Who is responsible to do this? This generates a circular ecosystem map that captures the circularity potential of your business context. For more detailed instructions on this workshop method, please read:

Konietzko, J.; Bocken, N.; Hultink, E.J. A Tool to Analyze, Ideate and Develop Circular Innovation Ecosystems. Sustainability 2020, 12, 417.

Narrow



**Design with
low-impact
inputs**

*Product
principle*

**Design products with
'ingredients' and materials
that require less land,
energy, water and/or
materials to produce.**

Example

**The company Impossible Foods
has designed a 'meaty'
plant-based burger. Compared
to the beef version, it requires
ca. 7 m² less land, 300l less
water and 5kg less CO₂ than
the meat-based alternative.**

Narrow



**Design
light-weight
products**

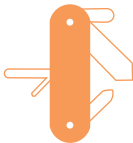
*Product
principle*

Design products that are lighter than comparable products on the market. Lighter products usually require less materials and need less energy to transport.

Example

Adaptive City Mobility has designed a car that weighs 650kg incl. battery. This is around 1/3 of an average car (in 2019), leading to ca. 50% less battery to operate.

Narrow



Design for multiple functions

*Product
principle*

Design products with multiple functions. Multi-functional products can reduce the overall number of products and may be usable by different user groups.

Example

Studio Davero has designed Puzzle Peace: a modular furniture kit that can be turned into multiple types of furniture; e.g., a bench, table, armchair, chair, bar or display.

Narrow



**Enable and
incentivize
users to
consume less**

*Business
model
principle*

Incentivize users to use less energy or material during the use of energy or material-using goods like washing machines or cars.

Example

HOMIE offers washing machines through a pay-per-wash model. By monitoring user behavior, the company increases the resource efficiency of laundry.

Narrow



**Organize
light-weight
urban
transport**

*Business
model
principle*

Organize lighter forms of transportation, for example electric tricycles. The lighter the vehicles, the lower the amount of energy and materials required to transport people and goods.

Example

RYTLE provides a cargo solution that consists of a light-weight vehicle, an easily exchangeable box, storage hubs and software that connects all entities.

Narrow



**Localize
supply where
appropriate**

*Business
model
principle*

Find more local suppliers, where appropriate. More local suppliers decrease the amount of energy needed to transport goods.

Example

The restaurant chain 'Dig Inn' has developed a supply system of local farmers in every region it operates in to provide a scalable, locally adapted and seasonal restaurant model, reducing travel distances for food ingredients and stimulating the regional economy.

Narrow



**Maximize
capacity use
of products**

*Ecosystem
principle*

Maximize the degree to which the capacity of a product is used. This is sometimes referred to as ‘sharing’, where multiple user groups have access to the same product. This can decrease the overall number of products in an ecosystem.

Example

The online platform Peerby enables people to share everyday goods like drills or bicycles, which can increase their usage and reduce the overall number of personally owned goods in homes over time.

Slow



Design for physical durability

*Product
principle*

Design products that degrade more slowly than comparable products on the market.

Example

A cast-iron pan can last much longer than comparable pan types.

Slow



Design for emotional durability

*Product
principle*

Design products that users will love and trust over a long period of time.

Example

The ‘Leatherman’, a pocket-knife sized toolkit, has a 25-year warranty and many uses. It lets people collect and experience personal stories and creates an emotional bond between the user and the product.

Slow



Design for ease of maintenance and repair

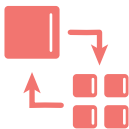
*Product
principle*

Design products that can be easily maintained or repaired. Maintaining means inspecting the product to retain its functional capabilities. Repairing is about restoring a product to a sound/ good condition after decay or damage.

Example

Fairphone has designed a modular phone that can be easily disassembled to repair and exchange components.

Slow



Design for easy dis- and reassembly

*Product
principle*

Design products that can be easily separated and reassembled.

Example

Gerrard Street has designed a pair of headphones that can be easily separated and reassembled.

Slow



Design for upgradability

*Product
principle*

A product is upgradable if its functionality or performance can be improved during or after use.

Example

An example is a bicycle with exchangeable and upgradable components.

Slow



Design for standardization and compatibility

*Product
principle*

**Create products,
components or interfaces
that also fit other
products, components or
interfaces.**

Example

**A common example of a
standardized component is the
Mini-USB slot for charging.**

Slow



**Enable users to
maintain and
repair their
products**

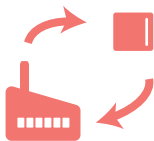
*Business
model
principle*

Create services that enable users to care for their product.

Example

Fairphone supports users to care for their phones through discussion forums on how to maintain and repair, and an inventory of spare parts.

Slow



**Remanufacture
existing
products and
components**

*Business
model
principle*

Recover value from collected end-of-use products by reusing their components for the manufacturing of products with the same functionality.

Example

The Chinese company Guangzhou Huadu collects used vehicle parts and remanufactures them into as-new certified spare parts.

Slow



**Repurpose
existing
products and
components**

*Business
model
principle*

Take existing products and components and take them out of their context to create new value with them.

Example

Ubitricity turns lamp lanterns in cities into charging stations for electric vehicles.

Slow



**Provide an
unconditional
lifetime
warranty**

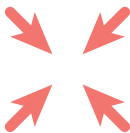
*Business
model
principle*

Offer your customers a life-time warranty, to add a promise that your products are made to last.

Example

Darn Tough Vermont produces socks with an unconditional lifetime guarantee. They say: “If our socks are not the most comfortable, durable and best fitting socks you have ever owned, return them for another pair.”

Slow



Encourage sufficiency

*Business
model
principle*

Encourage your customers to moderate the consumption of your products.

Example

In 2011, the clothing company Patagonia advocated: “Don’t buy this jacket.” With Worn Wear, it opened an online store for used Patagonia clothing, encouraging users to maintain their gear and trade it back once they don’t use it anymore.

Slow



**Provide the
product as a
service**

*Business
model
principle*

Offering the product as a service keeps the ownership with the firm and creates incentives to increase their lifetimes. You can offer product-, use-, or results-oriented models.

Example

The company Kaer offers a result: cool and fresh air as a service, rather than air conditioners as products.

Slow



Organize maintenance and repair services

*Business
model
principle*

Make sure that your products can last longer through maintenance and repair services. They can be offered by the manufacturer of a product or by third-party providers.

Example

The company Nudie Jeans, at its point of sale, has started to offer a free repair service for their Jeans. This has changed the shop appearance into a craftsmanship atmosphere.

Slow



**Provide
services that
upgrade or adapt
your products**

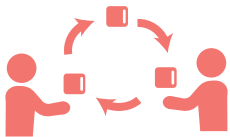
*Business
model
principle*

A product is upgradable if its functionality or performance can be improved during or after use. Try and integrate upgrading services into your offering.

Example

Gispen offers REMADE, a service to repurpose old furniture to fit new trends and work space requirements.

Slow



**Turn disposables
into a reusable
service
ecosystem**

*Ecosystem
principle*

Make use of or provide services that replace disposable with durable products.

Example

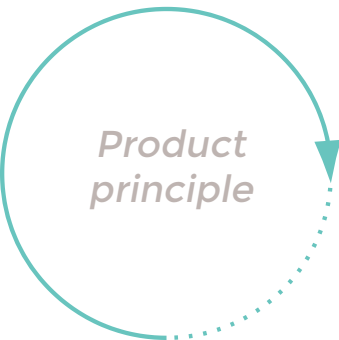
TerraCycle has designed ‘Loop’, which delivers products like shampoo or ice cream in reusable packaging. The packaging gets picked up, cleaned and used again. Involved actors include retail brands, service providers (e.g. cleaning and transport service) and end users.

Close



Design with recycled inputs

*Product
principle*

A circular arrow diagram. A solid teal line forms a circle, with a teal arrowhead pointing downwards on the right side. The bottom portion of the circle is represented by a dotted teal line.

Design with materials that have been recycled from other products and components.

Example

The ‘Design for Recycled Content Guide’ supports firms in opting for more recycled content in their products.

Close



**Design components,
where appropriate,
with one material**

*Product
principle*

A circular arrow diagram. A solid teal line forms a circle, with a teal arrowhead pointing downwards on the right side. The bottom portion of the circle is represented by a dotted teal line.

Composite materials are often hard to recycle, because they cannot be separated. Design components therefore, where appropriate, with only one material to increase recyclability.

Example

Adidas has launched Futurecraft.Loop, a shoe made from one recyclable material and no glue. It can be recycled into pellets that can be turned into a new shoe.

Close



**Design with
materials
suitable for
primary recycling**

*Product
principle*

Design for recycling that can turn end-of-life materials into new materials with equivalent properties.

Example

Aquafil has designed the Econyl system, which enables Polyamide 6 or Nylon 6 waste to be manufactured into new Nylon 6, with no loss of quality.

Close



**Reuse and sell
components and
materials from
discarded products**

*Business
model
principle*

**Create new value from
wasted products and
components.**

Example

**Roetz recovers bicycle
components that were thrown
away, and uses them to let
customers assemble their own
bikes.**

Close



Enable and incentivize product returns

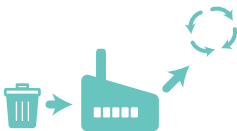
*Business
model
principle*

Make sure you can get back the products that you put on the market.

Example

Teemill makes users send back old and worn out products. Users can scan a QR code in the wash-care label to generate a free post label, which can be used to send the garment back to Teemill. Sending back products earns users credit for their next purchase.

Close



**Recycle products
in proper facilities**

*Business
model
principle*

Make sure that the products you put on the market get recycled in proper facilities.

Example

The initiative ‘Closing the Loop’ supports users and sellers of phones to be material-neutral and waste free. It collects scrap phones on behalf of customers and recycles them.

Close



**Build local
waste-to-product
loops**

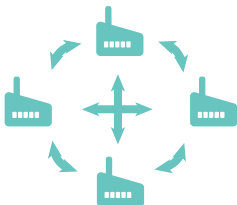
*Ecosystem
principle*

Create local resource loops by turning the waste of a given facility into new products that can be sold back to the facility.

Example

SOOP has formed an ecosystem of actors that collect coffee grounds and orange peels from offices, process them, make new products from them, and deliver these products to the same offices.

Close



**Engage in
industrial
symbiosis**

*Ecosystem
principle*

**Share or exchange
by-products, materials,
energy, or waste among
nearby firms.**

Example

**The Kalundborg Eco-industrial
park exemplifies how firms can
collaborate to share by-
products from their factories.**

Regenerate



Design self-charging products

*Product
principle*



Design products that can charge themselves with renewable energy. This is especially relevant for mobility assets.

Example

The company Sono Motors has designed a car with solar cells integrated into its body, allowing it to charge itself throughout the day.

Regenerate



Design with renewable materials

*Product
principle*



Design products with renewable and low-carbon materials. Renewable materials should only be chosen when its extraction rate is equal to or lower than its recovery rate. Further, next to its properties, materials need to be selected based on their expected end-of-life treatment to avoid unintended consequences.

Example

Ecovative produces mycelium-based fibres and materials with natural glue properties. Used for packaging, apparel and in the medical industries.

Regenerate



**Design with
non-toxic
materials**

*Product
principle*



Avoid using toxic materials and substances in any of your products or operations. Toxic substances tend to accumulate in the biosphere and cause negative health effects for humans and other species.

Example

Vestaron substitutes synthetic pesticides with biological ones that are safe for humans, birds, fish and pollinators.

Regenerate



**Produce and
process with
renewable
energy**

*Business
model
principle*



Build up your capacity as a company to produce and process with renewable energy.

Example

In 2019, Apple has been the company with the biggest installed capacity for solar energy, 400 MW.

Regenerate



Power transportation with renewable energy

*Business
model
principle*



Find ways of how you can power your transportation needs with renewable energy

Example

The company Foodlogica links local food, consumers and businesses in Amsterdam's city center through a light-weight mobility system, powered by renewable energy.

Regenerate



**Power the use of
the product with
renewable
energy**

*Business
model
principle*



Find ways of powering your product with renewable energy, through creative partnerships or product and service design.

Example

Waka Waka provides portable devices with photovoltaic panels that can power every-day electronics.

Regenerate



**Embed renewable
energy production
in the existing
infrastructure**

*Ecosystem
principle*



Find ways of making renewable energy production part of the existing infrastructure.

Example

‘Solar Roadways’ has developed a modular system of solar panels that can be walked and driven upon.

Regenerate



**Recover nutrients
from urban areas**

*Ecosystem
principle*



Find ways of recovering valuable nutrients from urban areas that are usually lost.

Example

Lystec Inc. helps the city of Guelph to turn biosolids from wastewater treatment into organic nutrients for surrounding agriculture.

Regenerate



Regenerate polluted ecosystems

*Ecosystem
principle*



Contribute to regenerating polluted ecosystems that affect your business.

Example

The Ocean Cleanup Project develops technology to clean oceans from plastic pollution.

Regenerate



Manage and sustain critical ecosystem services

*Ecosystem
principle*



Engage in projects that manage and sustain the natural ecosystems that surround and affect your business operations.

Example

Nestle's Häagen-Dazs has partnered with a nonprofit to plant the largest pollinator habitat in the United States. Without pollination services from bees, many critical ingredients for the ice-cream would not exist anymore.

Inform



**Use artificial
intelligence to
develop new
materials with
circular properties**

*Product
principle*

Developing a new material requires data about the structure and properties of materials. AI can help analyze the required and available data quickly to inform design decisions based on circular requirements.

Example

The ‘Accelerated Metallurgy project’, run by the European Space Agency, has used AI to produce and test new metal alloys. AI helped the project to speed up the search for new materials.

Inform



Virtualize

*Product
principle*

**Deliver utility virtually.
Virtualizing reduces the
need for materials to deliver
the same utility.**

Example

**reMarkable has designed a
digital device that “feels like
paper”. A tool for note-taking,
reading and reviewing
documents.**

Inform



Design connected products

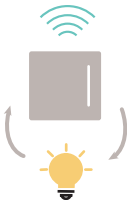
*Product
principle*

Design products with sensors that can send and receive different types of data.

Example

Delta Development, as part of its product-as-a-service model, has sensors in some of their elevators to inform maintenance needs.

Inform



Use product-in-use data to design more circular products and services

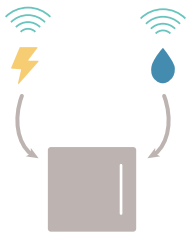
Business model principle

Data on how a product is used can be valuable to make better design decisions for future products and services.

Example

Rolls-Royce aggregates product-in-use data to make their engines more efficient, durable and long-lasting.

Inform



**Track the resource
intensity of the
product-in-use**

*Business model
principle*

Tracking the resource use of products while they are used can, for example, help to influence user behavior to decrease the resource intensity of product use.

Example

Philips uses sensors to track data on how their lights are used within their lighting-as-a-service model to save electricity.

Inform



**Track the
condition, location
and/or availability
of the product**

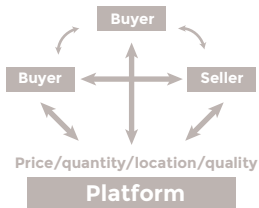
*Business model
principle*

Tracking the condition of the products can help to predict when it will be necessary to service (repair, maintain) them. Tracking the location and availability can, for example, enable the maximizing of their use capacity.

Example

Zipcar uses the data to optimize their car sharing service.

Inform



**Market circular
products,
components and
materials through
online platforms**

*Ecosystem
principle*

Online platforms can serve to market circular products, components and materials.

Example

Stuffstr buys and collects used products from consumers and sells them in second hand markets. An AI algorithm helps the company to set competitive prices for the seller, while ensuring a good margin on the second hand market.

Inform



Build material database ecosystems

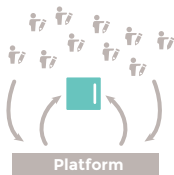
*Ecosystem
principle*

Create or leverage material databases. They describe the characteristics of materials and components in products so that products can be more easily reused and their materials recovered.

Example

The project ‘Buildings as Material Banks’ has brought together different stakeholders to develop a material database ecosystem for buildings.

Inform



**Co-create products,
components,
materials and
information via
online platforms**

*Ecosystem
principle*

Online platforms can be used to 'crowdsource' design projects for circular products, components and materials.

Example

The online platform launchforth.io connects designers and engineers with firms to co-create new products.

Inform



**Use artificial
intelligence to
optimize circular
infrastructure**

*Ecosystem
principle*

The circular economy requires the collecting, sorting, separating, treating, and redistributing of products, components and materials. Often, products, components and materials are diverse and difficult to handle. Artificial intelligence can help to optimize the infrastructure required for a circular economy.

Example

ZenRobotics uses robots with cameras and sensors to automatically sort all kinds of waste streams with an accuracy level of 98 %.

Inform



Platform

Operate service ecosystems via online platforms

*Ecosystem
principle*

Online platforms can serve to operate service ecosystems that require several actors who need to coordinate their interactions and economic exchange.

Example

The online platform Whim operates mobility-as-a-service ecosystems in cities that combine different private and public transportation options for a seamless mobility experience.